

AI Traffic & Parking Analyzer

Project Overview — Awotwi-Baf

1. Problem

Finding a free parking space wastes time, and reserved spaces (such as faculty parking) are often taken by the wrong vehicles. Most lots already have cameras, but the footage is only recorded, not analysed. This project turns an ordinary parking lot video into live occupancy and violation information.

2. Objectives

- Detect vehicles moving through the lot.
- Track each moving vehicle across frames with a persistent ID.
- Decide which of the 22 parking spaces are available or occupied.
- Flag wrong parking in reserved Faculty spaces.
- Show everything on a live dashboard.

3. Methodology

Spot setup: each space is drawn as a polygon on a video frame and labelled Faculty or Others; the polygons are saved as JSON.

Detection: MOG2 background subtraction finds moving vehicles, and HSV colour thresholding picks out the flagged (blue) car.

Tracking: a centroid tracker matches detections between frames to give each vehicle an ID and a path.

Occupancy: each space is compared with a photo of the empty lot. If at least 50% of the space's pixels differ, the space is occupied.

Wrong parking: a space is flagged when the flagged car covers at least 30% of a Faculty space.

Automatic spot detection: white parking lines can also be found with colour thresholding and morphology, and spaces are built between neighbouring lines.

4. Tools & Technologies

Python 3, OpenCV, NumPy, Streamlit.

5. Outputs

- Annotated video with detections, IDs and trajectories.
- Colour-coded parking plan (green = Faculty, blue = Others, yellow outline = occupied, blinking red = wrong parking).
- Live Available / Occupied / Wrong Parking / Total counts on the dashboard.

6. Future Work

Deep-learning vehicle detection (e.g. YOLO), licence plate recognition to identify authorised vehicles, occupancy history and analytics, and support for multiple cameras.

AI Traffic & Parking Analyzer — System Architecture

